

RESEARCH STATEMENT

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*When the Model Becomes the Risk: Scalable Bayesian Inference
for Trustworthy Environmental Health Surveillance*

My research interests span Bayesian methodology, scalable spatio-temporal modeling, and environmental health statistics. A common thread in my work is understanding the theory and design of statistical models that remain reliable when applied to data collected across many locations over many years. I draw on methods of inference from Bayesian statistics, spatial statistics, numerical linear algebra, and dimension reduction. Broadly speaking, my research belongs to the area of *scalable Bayesian spatio-temporal inference* — the study of how to build large hierarchical models that preserve the integrity of the scientific conclusions they produce — an area whose theoretical foundations, applied to national environmental surveillance, are only beginning to be laid.

Background and Current Work

Federal agencies build their understanding of environmental risk from data collected across thousands of locations over many years. Whether such a surveillance system functions depends on the statistical model used to separate the effect of an exposure from the background variation that surrounds it. Hierarchical Bayesian modeling is the standard technique for capturing this background variation. It would be naive, however, to expect that simply combining spatial and temporal random effects yields a reliable model. The performance of any such model is governed by two things: (1) the dimension of the basis used to represent spatio-temporal structure, and (2) the degree to which that structure competes with the exposure of interest for the same variation in the data.

There is a rich body of prior work in spatial statistics, Bayesian computation, and environmental epidemiology that answers many fundamental questions about hierarchical modeling and the design of spatio-temporal random effects. This work, however, has not been carried out at the scale of national surveillance. Pressed by the rapid growth of environmental and health data, both academic and agency practice have made do with over-parameterized or hand-tuned models, producing analyses that offer little guarantee regarding the integrity of the exposure estimate. Standard spatio-temporal models in use today can, for example, absorb signal that belongs to a chemical exposure and, in the worst case, reverse the apparent direction of its effect. It is easy to see how this can adversely affect the regulatory decisions that protect public health.

We need to understand the general principles for designing large hierarchical models that preserve the integrity of the conclusions agencies act on. Such guarantees may be statistical — for example, well-calibrated posterior uncertainty for an exposure effect, or structural — for example, an exposure surface estimated free of spatial confounding by construction. Only recently has interest grown in the theory and design of scalable, self-correcting Bayesian surveillance models. In my Ph.D. thesis, I lay a framework for this problem. Much of this work has benefited greatly from interaction with my advisor and colleagues. I describe it below.

An Analytical Framework to Compress Spatio-Temporal Models

Over the past decade, spatio-temporal modeling has enjoyed both wide academic interest and applied success. I ask the following question: *is there a common technique that lets us reduce a spatio-temporal model to its essential components while preserving valid inference?* In the course of solving two open problems about scaling spatio-temporal inference, I found such a technique, built on the singular value decomposition:

1. *Can we retain the accuracy of a full spatio-temporal model using only a small, automatically chosen set of components?*
2. *Can we do so without expert tuning of the model dimension by the analyst?*

SVD-PICAR is a compression of the projected intrinsic conditional autoregressive (PICAR) basis. I showed that spatio-temporal model design can be cast as the problem of reducing a large, strongly collinear basis to the few orthogonal directions that genuinely explain the outcome. That these problems admit such a simple solution through the singular value decomposition is striking, because it refutes the common assumption that scale must be traded against reliability [1, 2]. The analysis is illuminating because it isolates the structural requirements of a valid spatio-temporal model. I developed a generic framework for a class of models I call *compressed spatio-temporal models*, showing how the SVD yields well-calibrated inference for Gaussian and multinomial outcomes [1], and later extending the analysis to over-dispersed count outcomes within a kernel-machine regression [2]. This framework now covers nearly all of the outcome types in use in environmental surveillance today, and shows how spatio-temporal inference can be scaled efficiently.

Deterministic Separation of Signal from Background

Every environmental health analysis must separate an exposure effect from spatio-temporal background variation. This remains poorly understood, largely because of the variety of outcomes and exposures involved. These tasks place heavy demands on computation and posterior stability, which prevents them from being implemented reliably with off-the-shelf

hierarchical models. Specific solutions exist, but in most cases it is not known whether they are stable, whether they are complete (i.e., support all necessary outcome types), and whether they preserve the integrity of the exposure estimate. I examine three aspects of this problem.

1. *Scalable and stable spatio-temporal inference that eliminates redundancy.*

Spatio-temporal models are built by combining spatial and temporal random effects, which, as is well known, produces a strongly collinear basis and in turn unpredictable posterior behavior and degraded mixing. I instead proposed a deterministic compression that exploits the spectral structure of the spatio-temporal basis to retain only its informative directions. I showed how the optimal low-rank representation can be characterized through the singular value decomposition, and used adversarial simulation designs to show that it preserves accuracy [1]. The resulting model retains the fidelity of the full hierarchical model while attaining the speed and stability of a compact one, removing the need for hand-tuned dimension reduction. Several environmental and surveillance applications, including NASA satellite measurements and national cancer surveillance records, have demonstrated the value of this technique.

2. *Stable architectures for chemical mixture effects in count data.*

I showed that there is a direct relationship between compression of the spatio-temporal basis and recovery of an unbiased chemical mixture effect, and showed analytically how a kernel-machine regression that preserves the integrity of the exposure estimate can be designed for over-dispersed count outcomes [2].

3. *A complete and flexible framework for environmental outcomes.*

Surveillance requirements vary widely: cancer incidence requires count models, while exposure-classification tasks require multinomial models. Previous research has focused on modeling each outcome at high fidelity in isolation. I instead developed a framework that is both flexible and complete — one that applies across many outcome types while preserving valid inference. As a first step, we identified the elementary building blocks of compressed spatio-temporal inference in terms of the SVD basis, then designed a unified Bayesian framework spanning Gaussian, multinomial, and count outcomes [1, 2]. This foundation also supported a collaborative epidemiologic study linking drought to COPD mortality [3].

Confounding-Aware Inference for Environmental Exposures

Environmental epidemiology abounds with flexible models that fit the data well but absorb the exposure signal. What we want instead are models that fit the background while protecting the exposure estimate. I address two open questions that shed light on two classes of practical models.

1. *Is there a spatio-temporal model that preserves the exposure estimate by construction?*

The combined spatial-temporal random effect is the most common background structure in environmental surveillance, yet the flexibility needed to capture background variation tends to absorb the exposure signal and carries a high risk of confounding. I analyzed confounding-aware compression within the SVD-PICAR framework and derived analytically the conditions under which the compressed model yields estimates that are both well-calibrated and protected against confounding. Because the exposure surface can be projected onto the orthogonal complement of the confounding basis, it is estimated free of spatial confounding by construction. These results show that surveillance analyses built on standard spatio-temporal models can be upgraded in a practical way, using our compression to deliver reliable exposure estimates [2].

2. *When can compact models give optimal inferential performance?*

Contrary to intuition, a more flexible spatio-temporal model that maximizes in-sample fit may not preserve the validity of the exposure estimate, so over-flexible models are dangerous in practice. Compact models, by contrast, are attractive for their stability and speed, and I show the conditions under which such compressed models give accurate, well-calibrated inference [1].

Scalable Surveillance Inference — A Research Agenda

In the course of this research, I have found that the overhead in computation, complexity, and tuning of designing surveillance models that preserve the integrity of the exposure estimate is small, for two reasons. First, the inherent spectral structure of spatio-temporal bases makes many of these problems tractable. Second, advances in numerical linear algebra, insights from Bayesian computation and dimension reduction, and analysis techniques from spatial statistics together enable elegant and simple solutions. I envisage a field of *scalable Bayesian surveillance* built up from the foundations of these areas.

In the near term, I am pursuing three thrusts that move this framework from a dissertation method toward deployable federal surveillance infrastructure. *First*, spatially varying exposure effects: I will embed a spatially varying coefficient process within the compressed basis and derive the SVD-PICAR analogue of restricted spatial regression, so that the exposure surface is estimated free of spatial confounding by construction and agencies can map where an exposure does the most harm. *Second*, critical exposure windows: I will develop a distributed-lag formulation in which the temporal SVD directions play a dual role — capturing the background trend and spanning a penalized low-rank lag surface with a fused penalty enforcing toxicologically plausible smoothness, so that the specific exposure years driving an outcome can

be recovered with honest uncertainty. *Third*, automatic model selection: I will develop and theoretically characterize information-criterion and predictive procedures (WAIC, leave-one-out cross-validation, and an SVD marginal-likelihood criterion) that select model rank automatically, delivering both consistency guarantees and a reproducible pipeline an NCHS or EPA analyst can run without statistical adjudication.

Looking further ahead, although scalability will remain central, I also intend to study the *robustness to misspecification, graceful degradation, and reproducibility* of surveillance systems, which will only grow in importance. As these systems become larger and more interdependent, *simplicity of design and method re-use* will become major considerations, and dimension reduction can play a key role: many of the solutions I propose rely on re-use and compression, which serve exactly these goals.

My work will balance the forward-looking and the immediate. Part of it will pursue fundamentally different proposals and more radical solutions — for example, can we achieve trustworthy, fully automatic environmental surveillance at national scale, given a spatio-temporal model that protects the exposure estimate by construction? The other part will focus on practical systems of immediate relevance to agencies such as NIEHS, CDC-NCHS, NCI, and EPA. I intend to work closely with researchers in related fields and to collaborate with agency scientists on practical problems, drawing on my NASA ARSET certifications in satellite-derived environmental data. My experience conducting research jointly with colleagues will help me do this. I am excited at the prospect of learning, contributing, and making a lasting impact in this emerging and challenging field.

References

- [1] K. B. Afari, Y. Gwon, “Compressed Spatio-Temporal Basis Functions: An SVD-PICAR Framework for Scalable Gaussian and Multinomial Inference,” submitted to *Journal of Computational and Graphical Statistics*, 2026. Under review.
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